Exercise 04: Heap Sort

Write a text file named '`heap.txt`' that constructs a trace table that only tracks the changes made to its array parameter for each of the following callers. If the definition of the function caller invokes another function, all the changes made to the array parameter by the embedded caller must be treated as a single change.

Callers

1. `MAXHEAPIFY(A,3)` where `A = [12,81,23,43,12,90,72,19]`
2. `MAXHEAPIFY(A,0)` where `A = [18,80,34,66,49,55,90,27]`
3. `BUILDHEAP(A)` where `A = [32,75,21,50,63,10]`
4. `HEAPSORT(A)` where `A = [7,12,9,4]`